

SEID 2364 – Assignment 3 Harm Analysis

Part 1 – Speculative Harm Analysis

Selected System: AI Hiring System

An AI hiring system is used by companies to screen job applications and identify suitable candidates for interviews. The system collects applicant data such as resumes, educational background, work experience, and skills. Machine learning models may rank candidates automatically based on patterns from previous hiring decisions. Human recruiters then review recommendations produced by the system.

The main purpose of the system is to reduce hiring time and improve efficiency. However, the system can strongly influence who receives job opportunities. Because hiring decisions affect income, career growth, and social mobility, mistakes or bias within the system may create serious consequences for applicants.

Actions Performed Within the System

1. Collecting and processing applicant data 2. Ranking or filtering candidates automatically 3. Recommending applicants for interviews or rejection

Possible Harms

One possible harm is discrimination. If the AI system is trained using biased historical hiring data, it may unfairly favor certain genders, ethnic groups, or educational backgrounds while disadvantaging others.

Another possible harm is economic harm. Applicants who are unfairly rejected may lose employment opportunities and income because of inaccurate or biased system recommendations.

A third harm is reputational or psychological harm. Candidates may lose confidence in the hiring process if they feel the system is unfair or lacks transparency.

Part 2 – Readings Reflection

The readings helped me understand that harm is not limited to physical injury. Harm can also include social, economic, reputational, and psychological effects.

John Stuart Mill's Harm Principle suggests that actions should be limited when they cause harm to others. In the context of AI hiring systems, biased decisions may restrict opportunities and negatively affect people's lives.

The reading on moral theory explained that ethical reasoning often depends on structured frameworks used to compare harms and benefits. This is important because AI systems may increase efficiency for companies while simultaneously creating unfair outcomes for some applicants.

The consequentialism reading focused on evaluating actions based on their consequences. From this perspective, an AI hiring system may appear beneficial if it improves efficiency overall, but it becomes ethically problematic if the harms to disadvantaged groups outweigh the benefits.

Additional Research Reflection

One additional source I reviewed discussed algorithmic bias in hiring technologies. The source explained that automated systems can inherit historical bias from past company data. This reinforced the idea that AI systems are not neutral and can reproduce social inequalities if not carefully monitored.

Part 3 – Revised Harm Analysis

Types of Harm

1. Economic harm – unfair loss of job opportunities 2. Social harm – discrimination against certain groups 3. Psychological harm – stress, frustration, and loss of trust 4. Reputational harm – companies may lose public trust if bias is discovered

Entities Affected

1. Individual job applicants 2. Marginalized social groups 3. Companies using the system 4. Society as a whole

Scale of Harm

The harm may begin at the individual level, but if biased systems are widely adopted, the effects can spread across industries and society. Large numbers of people may face unfair barriers to employment opportunities.

Conclusion

This assignment showed me that harm within AI systems can exist even when the system is designed for efficiency or convenience. Harm analysis requires looking beyond technical performance and examining how decisions affect individuals and society. AI hiring systems may improve efficiency, but without transparency and fairness safeguards, they can also reinforce inequality and create serious ethical problems.